

Supplementary Material:

Exact Radial-Determinant Frequencies of a Piezoelectric Annular Plate, and Inverse Recovery of e_{31} from Open-Circuit Flexure

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Main text contains no job numbers; they are listed in §S.6.

S.1 Cubic consistency condition

Each layer carries $\varphi = \pm \bar{\varphi}f(s)$, $s = |z| - h \in [0, h_1]$, with $f(0) = f(h_1) = 0$ (mirrored so that the two layers' moments add). The thickness-integrated electric enthalpy of one layer is $-\bar{e}_{31}I_1\bar{\varphi}\nabla^2w - \frac{1}{2}\bar{\Xi}_{33}I_2\bar{\varphi}^2 - \frac{1}{2}\bar{\Xi}_{11}I_3|\nabla\bar{\varphi}|^2$ plus the elastic part, with $I_1 = \int z f' dz = -\int f ds$, $I_2 = \int f'^2 ds$, $I_3 = \int f^2 ds$. Stationarity in w and $\bar{\varphi}$ gives the flexural equation with piezo moment coefficient $K = -2\bar{e}_{31}I_1$ (two layers) and the electrical equation $\bar{\Xi}_{11}I_3\Delta\bar{\varphi} - \bar{\Xi}_{33}I_2\bar{\varphi} - \bar{e}_{31}I_1\Delta w = 0$; the insulated edge $\partial\bar{\varphi}/\partial r = 0$ is its natural condition. The long-wave increment per layer is $(\bar{e}_{31}I_1)^2/(\bar{\Xi}_{33}I_2) \leq h_1^3\bar{e}_{31}^2/(12\bar{\Xi}_{33})$ by Cauchy–Schwarz, with equality only for f' linear in s , i.e. the quadratic bubble $f = 4s(h_1 - s)/h_1^2$ of the main text: $I_1 = -2h_1/3$, $I_2 = 16/(3h_1)$, $I_3 = 8h_1/15$, so $K = \frac{4}{3}h_1\bar{e}_{31}$ and the normalized electrical equation is $a\Delta\bar{\varphi} - \bar{\varphi} + c\Delta w = 0$ with $a = h_1^2\bar{\Xi}_{11}/(10\bar{\Xi}_{33})$, $c = h_1^2\bar{e}_{31}/(8\bar{\Xi}_{33})$. The sine of Duan, Quek and Wang [1] gives $K = (4/\pi)h_1\bar{e}_{31}$; under this projection its long-wave increment is $96/\pi^4$ of exact, and under their thickness-integrated Gauss-law closure ($a = h_1^2\bar{\Xi}_{11}/(\pi^2\bar{\Xi}_{33})$, $c = h_1^2\bar{e}_{31}/(2\pi\bar{\Xi}_{33})$) it is $12/\pi^2$ of exact. The earlier version of this paper used that closure; the solver keeps it as a switch (`projection='duan'`), and it reproduces every earlier number bit-for-bit. Positing $\bar{\varphi} = \chi\bar{w}$ with χ independent of r gives $\lambda = \chi/(a\chi + c)$ from the electrical equation and $(d_1 + d_2)\lambda^2 + K\chi\lambda - A_2\omega^2 = 0$ from the flexural one; eliminating λ gives the cubic $Ka\chi^3 + (d_1 + d_2 + Kc - A_2\omega^2a^2)\chi^2 - 2A_2\omega^2ac\chi - A_2\omega^2c^2 = 0$ (Duan, Quek and Wang's Eq. 17a is the same cubic up to a constant factor, with their coefficients). Each of the three roots (χ_i, λ_i) depends on the trial frequency ω . For each root the radial problem is $\Delta\bar{w} - \lambda_i\bar{w} = 0$, with Bessel pair (J_n, Y_n) or modified pair (I_n, K_n) according to the sign of λ_i . The general short-circuit solution is the sum over all three branches. The limit $\bar{e}_{31} \rightarrow 0$ is singular (leading and two subleading coefficients of the cubic vanish together), so the elastic bilayer is a separate 4×4 .

The open-circuit rank-1 coefficient of the main text,

$$\alpha = \frac{4\bar{e}_{31}^2(h + h_1/2)^2h_1}{\bar{\Xi}_{33}(r_o^2 - r_i^2)}, \quad (\text{S.1})$$

is the unique sign that raises the first free-free $n = 0$ root. It is the product of the moment arm of M_p and the charge arm of Q , both $h + h_1/2$ (the bus charge is $-\partial\mathcal{H}/\partial V$ of the same enthalpy); the earlier outer-face charge arm $h + h_1$ made α too large by $(h + h_1)/(h + h_1/2)$. On a mixed edge the same αS is added only where $M_{rr} = 0$ is imposed (the free edge). Driven Q/V uses the same 4×4 with right-hand side $M_p = 2\bar{e}_{31}(h + h_1/2)V$ on both moment rows. An independent bisection of this Q/V recovers the open-circuit root to 1.4×10^{-15} relative.

S.2 Inverse on-branch cross-check

For the two clamped-clamped short-circuit recoveries, the product $(d\omega/dp)(dp/d\omega)$ on the solution branch is 1.000009 (e_{31}) and 1.000008 (E). Propagated uncertainties at 0.10% frequency precision are ten times the 0.01% values in main-text Table 4: 0.246% of E , 8885% of short-circuit e_{31} , and 21.3% of open-circuit e_{31} at $h_1/2h = 1/12$.

The sequential recovery (main text §6, job 2494216) was run at full production precision (dps=100) on the cluster and reproduces the in-sandbox validation run bit-for-bit. Round trip: $\hat{E} = 2.000000009 \times 10^{11}$ Pa (4.5×10^{-9} relative), then, holding $\hat{E}$, $\hat{e}_{31} = 4.100001626 \text{ C m}^{-2}$ (4.0×10^{-7} relative); both below the 10^{-6} bar. On-branch sensitivities at $h_1/2h = 1/12$: $d\omega_{\text{SC}}/dE = 5.678 \times 10^{-9} \text{ rad s}^{-1} \text{ Pa}^{-1}$, $d\omega_{\text{OC}}/de_{31} = -0.8577 \text{ rad s}^{-1} \text{ m}^2 \text{ C}^{-1}$, $d\omega_{\text{OC}}/dE = 1.518 \times 10^{-9} \text{ rad s}^{-1} \text{ Pa}^{-1}$, giving an implicit $de_{31}/dE|_{\omega_{\text{OC}}} = 1.769 \times 10^{-9} \text{ C m}^{-2} \text{ Pa}^{-1}$. At 0.01% frequency precision this splits the 3.01% total of main-text Table 4 into a 2.133% direct term and a 2.122% term from propagating $\hat{E}$'s own 0.0246% uncertainty; at 0.10% frequency precision the same split gives 21.33% and 21.22%, total 30.09%. The negative control substitutes the clamped-clamped open-circuit root for the free-free one in the second step: $\hat{e}_{31} = 4.100662522 \text{ C m}^{-2}$ still solves (sign flip present), but $d\omega_{\text{SC}}/de_{31}|_{\hat{E}} = -7.664 \times 10^{-3} \text{ rad s}^{-1} \text{ m}^2 \text{ C}^{-1}$ propagates to 888.5% of e_{31} at 0.01% frequency precision – the same order as the single-parameter clamped-clamped figure in main-text Table 4, confirming the control fails as required.

S.3 Finite-element discriminator and mixed electrical boundary

Axisymmetric PLANE223, KEYOPT(1)=1001, millimetre-newton-second-tonne, relative permittivity, one mesh for every electrical condition compared. Mode 1 is the numerical rigid body (≤ 0.003 Hz). Mode 2 is bending: transverse displacement same sign and magnitude at the host top and bottom at $r = r_o$; radial displacement opposite. The ratio of those components is 19.71 (both-faces short-circuit), 19.75 (open-circuit), and 19.72 (outer-only ground). Open-circuit mechanical amplitudes are smaller under unity-normalization because VOLT participates in the norm; the ratio is the identifier.

The outer-only-ground deck (VOLT= 0 on $z = \pm(h + h_1)$, interface free) gives 119.242 Hz. That is 0.36% above true both-faces short-circuit (118.811 Hz) and is not used as the short-circuit baseline in the main text. Coupling-off is 118.810 Hz.

The same fair open-circuit/both-faces-short-circuit pair was cloned, unmeshed, to $h_1/2h = 1/8$ and $1/5$ (job 2494226). At $1/8$: mode 2 is 121.839 Hz open-circuit, 121.348 Hz short-circuit, discriminator ratio 19.77 and 19.70. At $1/5$: mode 2 is 127.809 Hz open-circuit, 127.047 Hz short-circuit, discriminator ratio 19.80 and 19.69. All four runs show mode 1 at 0 Hz (rigid body) and zero MAPDL *** ERROR *** lines in the MAPDL statistics block. The queue script's own automated pass/fail scan reported all four as FAILED; that scan greps each run's transcript for the literal string *** ERROR *** and matched it against that same string appearing inside this deck family's own pre-registration comment, not against an actual MAPDL error line. The raw NUMBER OF ERROR

MESSAGES ENCOUNTERED line in each transcript reads zero, and the run completed (RUN COMPLETED) in every case; the numbers above are read from that completed output, not from the scan's own label.

The same fair pair was then remeshed with the piezo through-thickness division count doubled from 2 to 4, leaving radial divisions, host through-thickness divisions, units, and electrical BC unchanged (job 2494665). Mode 2 frequencies are unchanged at the printed precision: 119.143 vs 118.811 Hz at $1/12$, 121.839 vs 121.348 Hz at $1/8$, 127.809 vs 127.047 Hz at $1/5$. Discriminator ratios remain 19.7–19.8. The 93%/90%/86% recovery of the earlier, non-reciprocal closed-form split was therefore not a through-thickness mesh effect; it is the charge-arm ratio $(h + h_1/2)/(h + h_1) = 0.929/0.900/0.857$. Against the reciprocal closed form the three finite-element splits are 99.9%/99.9%/99.8% of the analytic value.

S.4 Wide-axis admittance

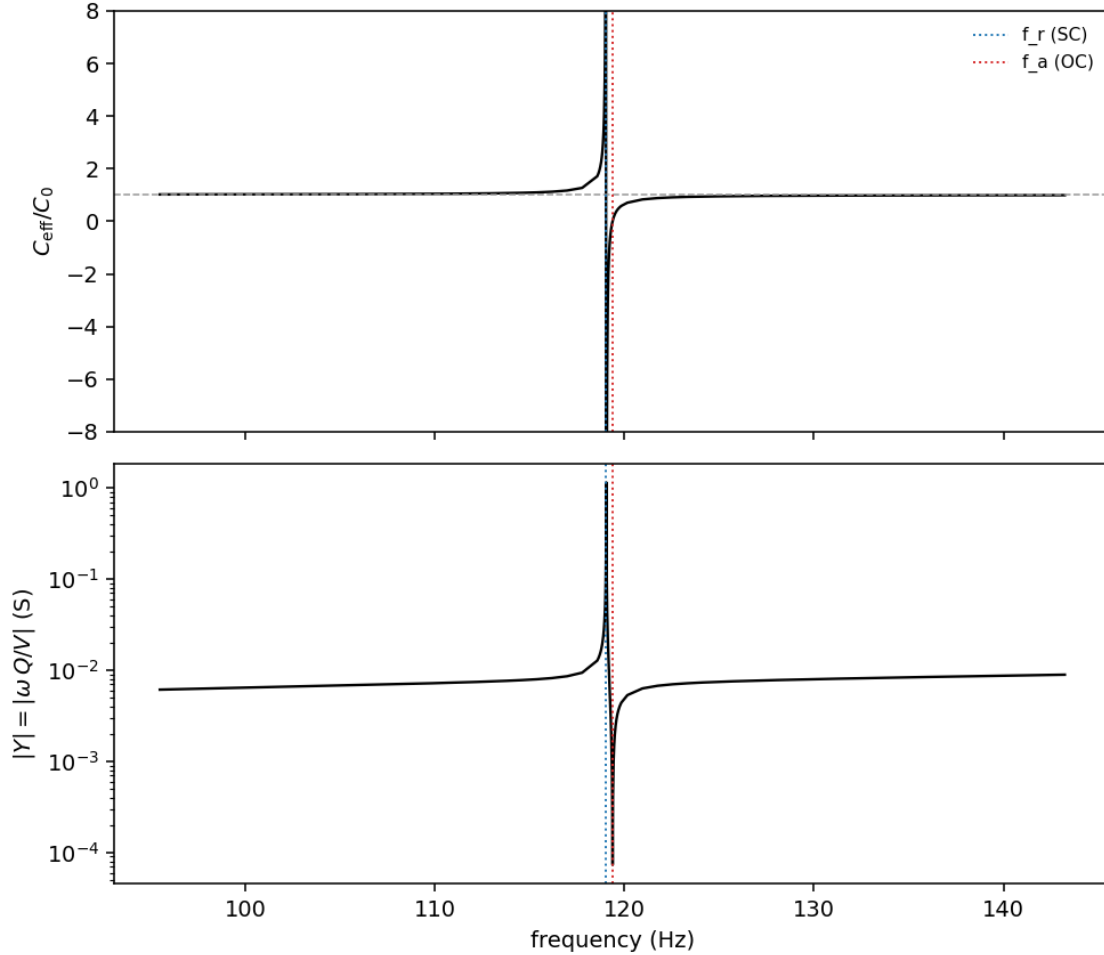


Figure S.1: Driven C_{eff}/C_0 and $|Y|$ on 95–145 Hz at $h_1/2h = 1/12$. The pole and zero of the first $n = 0$ pair differ by 0.28% and overlap on this axis; main-text Fig. 4 zooms 118.6–120 Hz.

A harmonic PLANE223 run on the same mesh (job 2494863; 1 V on the outer-bus master, KEY-OPT(1)=1001 charge reaction) places the $|Y|$ peak at 118.79 Hz and the charge-zero at 119.13 Hz,

against the modal pair 118.811 / 119.143 Hz. An earlier harmonic dump (job 2494669) used the current label **AMPS**, which is **KEYOPT(1)=101**, and returned a numerical-zero bus current at every frequency; that run is not a disproof of the pole/zero.

S.5 Thickness-ratio sweep and other-ratio admittance

Figure S.2 is the closed-form $h_1/2h$ sweep that sits underneath main-text Table 2 and Fig. 2: eleven ratios from 1/40 to 1/4, linear-potential open-circuit 4×4 against the elastic bilayer. The three Duan Table 4 ratios (red) lie on the curve. The thin-layer intercept $\Delta_{\text{OC}}/(h_1/2h) \rightarrow 0.035$ is the leading h_1 scaling of the rank-1 coefficient α (the same limit under either charge arm, since $h_1 \rightarrow 0$). The bottom panel is the single-parameter e_{31} G2 at 0.01% frequency precision (host E known exactly); it is 6.80% at 1/40 and 0.85% at 1/4.

Figure S.3 is the driven Q/V problem at the other two Table 4 ratios. Independent bisection of Q/V recovers `oc_ff_bisect` to 2.5×10^{-14} (1/8) and 9.6×10^{-14} (1/5); k_{eff}^2 from the driven pole/zero is 0.00805 and 0.01191. $C_0 = 6.66 \mu\text{F}$ at 1/8 and $4.16 \mu\text{F}$ at 1/5. Off-resonance Butterworth–van Dyke shape: $C_{\text{eff}}/C_0 = 1.019$ below the pole and 0.961 above the zero at 1/8; 1.028 and 0.943 at 1/5.

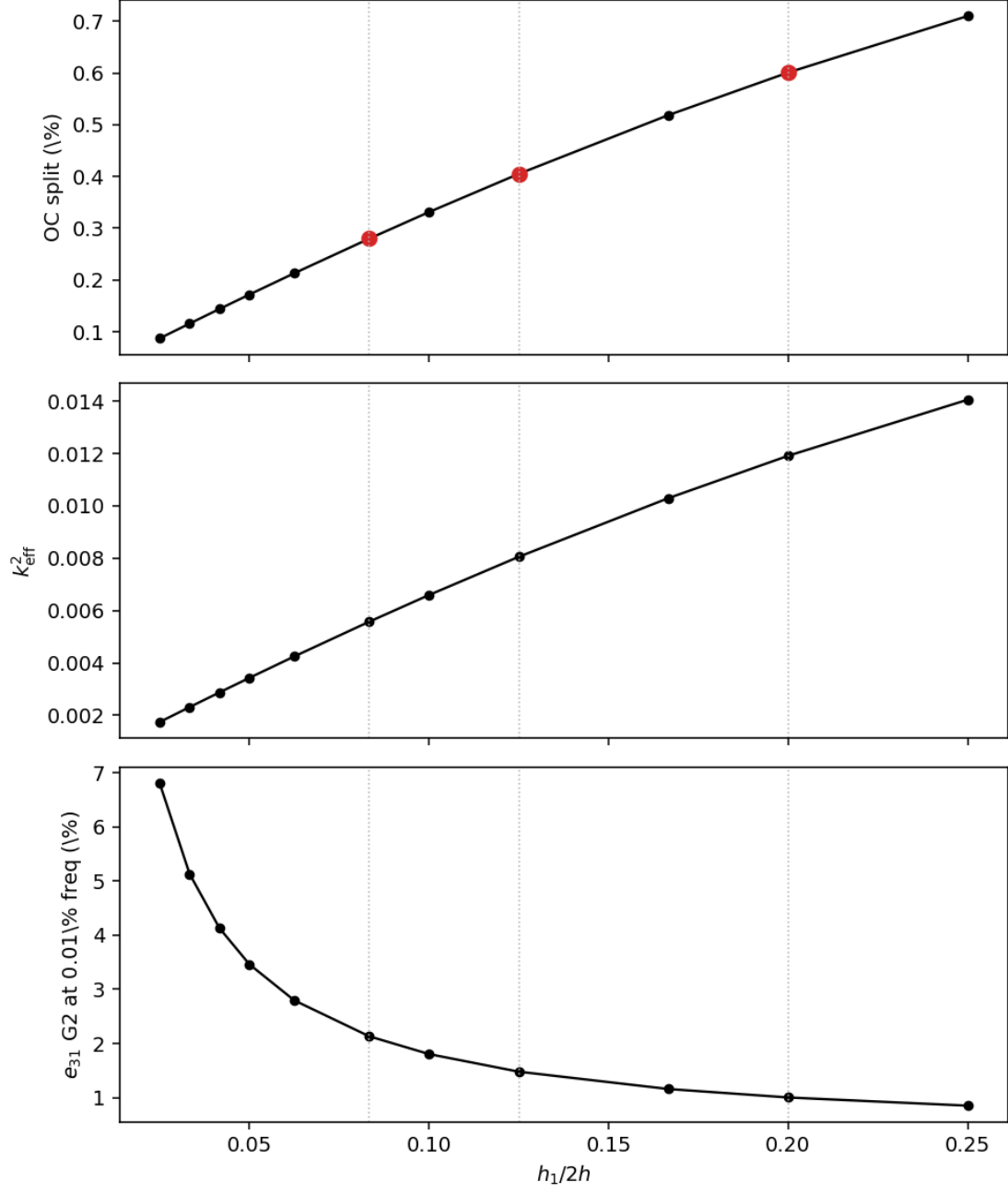


Figure S.2: Closed-form F-F $n = 0$ open-circuit split, k_{eff}^2 , and e_{31} G2 versus $h_1/2h$. Red markers are the three Duan Table 4 ratios already in the main text.

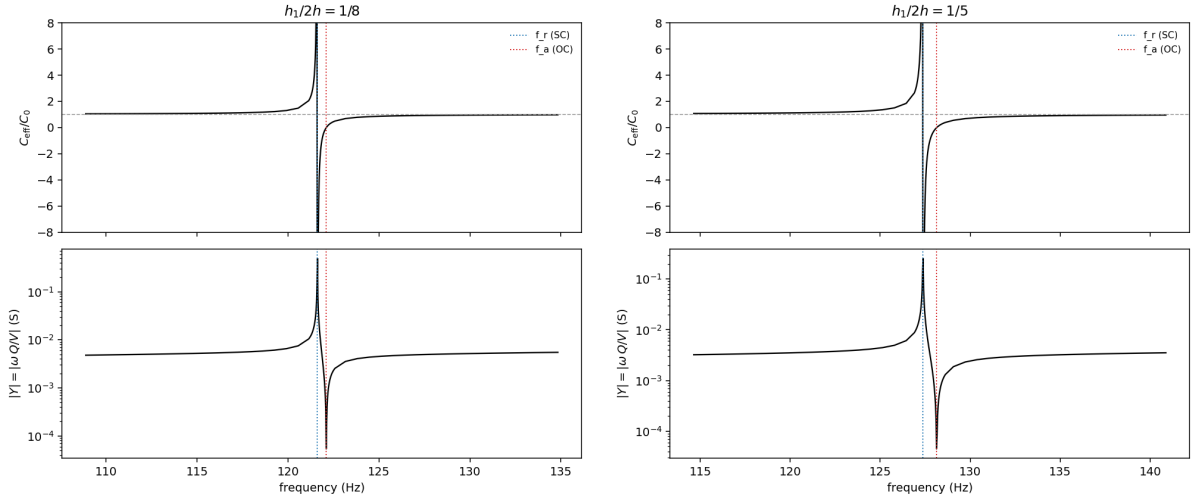


Figure S.3: Driven C_{eff}/C_0 and $|Y|$ at $h_1/2h = 1/8$ (left) and $1/5$ (right). Pole at the elastic/short-circuit root, zero at the open-circuit root, same linear-potential operator as main-text Fig. 4.

S.6 Cluster job provenance

All jobs on The Mill [3], solver snapshot 2026-07-10.s10. Piezoelectric paths are new and unused by the elastic default.

- 2489140 — clamped-clamped Table 4 gate, 27/27, worst -0.0476% .
- 2490706 — F-F coupling-off and outer-only-ground decks at $h_1/2h = 1/12$.
- 2490848 — F-F outer-only-ground at $h_1/2h = 1/8$ and $1/5$.
- 2491531 — clamped-clamped short-circuit inverse of E and e_{31} .
- 2491661 — F-F open-circuit python archival, deploy miss (package lacked the open-circuit method); superseded by 2491764.
- 2491761 — F-F open-circuit finite elements, mode 2 = 119.143 Hz.
- 2491764 — F-F open-circuit inverse archival, all gates passed.
- 2491831 — both-faces short-circuit finite elements, mode 2 = 118.811 Hz.
- 2491920 — driven $Y(\omega)$ archival, all gates passed, $k_{\text{eff}}^2 = 0.005997133$.
- 2494216 — sequential E -then- e_{31} inverse archival at full `dps=100`, $h_1/2h = 1/12$; SENTINEL PASS_ALL, bit-identical to the in-sandbox validation run.
- 2494226 — F-F open-circuit and both-faces short-circuit finite elements at $h_1/2h = 1/8$ and $1/5$, cloned from 2491761/2491831.
- 2494665 — lever 8 mesh refinement, NDIV_ZP $2 \rightarrow 4$ on the fair OC/scboth pair at all three Table 4 ratios; splits unchanged.
- 2494667 — lever 2 $h_1/2h$ sweep archival, SENTINEL PASS_ALL, bit-identical to the in-sandbox run.
- 2494668 — lever 5 driven $Y(\omega)$ at $1/8$ and $1/5$ archival, SENTINEL PASS_ALL, bit-identical to the in-sandbox run.
- 2494669 — lever 9 first harmonic dump; MAPDL clean but `RF,AMPS` identically zero. Superseded by 2494863.
- 2494863 — lever 9 v2 harmonic Y at $1/12$, `RF,CHRG`; pole 118.79 Hz, zero 119.13 Hz.
- 2494864 — lever 3 other-constant inverses, SENTINEL PASS_ALL.
- 2494865 — lever 1 $n = 0$ overtones; one overtone in $(800, 4000) \text{ rad s}^{-1}$ ($+0.137\%$ split). Pre-registered bar of two overtones in that window not met.
- 2494945 — lever 1 wide scan $(800, 15000) \text{ rad s}^{-1}$, SENTINEL PASS_ALL; second overtone 8515.06 vs 8516.48 rad s^{-1} ($+0.017\%$).
- 2494866 — lever 4 F-F E channel, SENTINEL PASS_ALL, 0.025% of E at 0.01% freq on both elastic and OC.

- 2495528 — lever 10 C–F / F–C mixed-edge archival, SENTINEL PASS_ALL, bit-identical to the in-sandbox `dps=60` run.
- 2495593 — optional C–F FE, clones of 2491761/2491831 with inner $U_X = U_Y = 0$ only. scboth 67.071 Hz, OC 67.245 Hz, +0.259% (92% of closed-form +0.282%). Both RUN COMPLETED, 0 MAPDL errors.
- 2495751 — lever 16 frequency-precision sweep, SENTINEL PASS_ALL, bit-identical to the in-sandbox run (F–F/C–F e_{31} uncertainty curve plus the Boeringa–McCune lab floor).

Consistent-projection rerun (2026-09-23). Every analytic job above (all but the finite-element jobs) used the Gauss-law closure of §S.1 and the outer-face charge arm. Every package-based probe was re-run unchanged against the corrected solver (default `projection='consistent'`; `projection='duan'` reproduces the jobs above bit-for-bit), with only the hard-coded open-circuit regression anchors of four probes moved to the new values. All pass (SENTINEL PASS_ALL), and every analytic number in this document and the main text is from that rerun. The rerun was done in-sandbox at the probes' production precision; it was not repeated on the cluster, because every earlier cluster archival run of these same probes reproduced its in-sandbox run bit-for-bit. The finite-element jobs are unchanged.

S.7 Mixed-edge C–F / F–C closed form

Paper 3 names inner then outer: C–F is clamped inner, free outer (`bc_inner='C'`, `bc_outer='F'` in `ring_disk.py`). Methods on `PiezoOutOfPlaneSolver`: `elastic_cf_det` / `elastic_fc_det`, `cf_coupled_det` / `fc_coupled_det`, `oc_cf_det` / `oc_fc_det`. The mixed-edge engine recovers the existing free–free and clamped–clamped determinants bit-exactly when both edges are set equal. Bare-host $n = 0$ roots match Paper 3's live Frobenius solver at relative 4.45×10^{-10} (C–F, 410.074181 rad s^{−1}) and 2.28×10^{-10} (F–C, 873.221632 rad s^{−1}). In-sandbox SENTINEL PASS_ALL (`probe_piezo_p4_cf_mixed_edge_2026-09-16.py`, `dps=60`, 2026-09-23 consistent-projection rerun): C–F open-circuit +0.262%, $k_{\text{eff}}^2 = 0.00522$, e_{31} G2 2.28% at 0.01% frequency; F–C open-circuit +0.00538%, $k_{\text{eff}}^2 = 0.000108$. Inverse round-trip on C–F e_{31} is 4.0×10^{-9} . (Cluster archival job 2495528 confirmed the earlier closure's +0.282% bit-for-bit.) Optional C–F FE job 2495593: scboth 67.071 Hz, OC 67.245 Hz, split +0.259% (98.9% of +0.262%; 92% of the earlier +0.282%), no 0 Hz rigid body, bending discriminator, 0 MAPDL errors. SOLVER_VERSION remains 2026-07-10.s10.

S.8 Frequency-precision sweep and lab noise floor

Table S.1 extends the two-column (0.01%, 0.1%) e_{31} inverse of the main-text open-circuit section and §S.7 to an eight-point log-spaced grid of assumed relative frequency error, on both the free–free and clamped–free $n = 0$ open-circuit channels, using the identical linear-propagation formula (unchanged, not retuned): $\delta e_{31}/e_{31} = (\delta\omega/\omega)\omega_{\text{target}}/(|d\omega/de_{31}|e_{31,\text{true}})$. The lab-floor row (marked) is sourced from Table 3 of [2]: an HP4294A impedance analyzer and a scanning laser Doppler vibrometer measured the same five axial resonances of a free–free PZT-4 rod specimen (a physically distinct specimen from this ring, same material system), disagreeing by 0.12%, 0.10%, 0.09%, 0.11%, and 0.09% – mean 0.102%. The companion draft (`BoeringaMcCune2026-2.pdf`) was checked and rejected as a source: its impedance-versus-SLDV table is internally inconsistent

Table S.1: Propagated relative uncertainty in e_{31} versus assumed relative frequency error, $h_1/2h = 1/12$. G0 round-trip error is 4.0×10^{-9} (bar 10^{-6}) at every row; G_anchor reproduces the 0.01% entries of main-text Table 4, and the 0.10%-scaled figures of §S.2, to within 1% relative; the linearity self-check (every row equal to the base row scaled by the ratio of assumed frequency errors) is exact to 10^{-9} .

assumed rel. freq. error	F–F OC unc. of e_{31}	C–F OC unc. of e_{31}
0.001%	0.213%	0.228%
0.003%	0.640%	0.683%
0.01%	2.133%	2.276%
0.03%	6.399%	6.829%
0.1%	21.331%	22.764%
0.102% (Boeringa–McCune floor [2])	21.758%	23.219%
0.3%	63.993%	68.291%
1%	213.311%	227.638%

(disagreements up to 26% on a single mode) and carries explicit placeholder text, so no number was taken from it.

Delivered as `Paper4_Piezo/probe_piezo_p4_lever16_freq_noise_floor_2026-09-16.py` (LF) with matching submit script (`dps=60`, mirroring the free–free/clamped–free probes’ own settings). In-sandbox SENTINEL PASS_ALL, ~ 85 s wall. Cluster archival job 2495751 (`compute-34-04.mill`, ~ 101 s wall, empty `.err`): SENTINEL PASS_ALL, every reported number bit-identical to the in-sandbox run (only per-call wall-clock timings differ). No package code was touched and SOLVER_VERSION remains 2026-07-10.s10.

References

- [1] W.H. Duan, S.T. Quek, Q. Wang, “Free vibration analysis of piezoelectric coupled thin and thick annular plate,” *J. Sound Vib.* 281 (2005) 119–139.
- [2] Z.L. Boeringa, G.H. McCune, D.S. Stutts, “Canonical validation and model discrimination in a linearly tapered non-uniform rod,” unpublished manuscript (2026).
- [3] Missouri University of Science and Technology, “The Mill HPC Cluster,” *The Mill* (2024), Article 1. <https://scholarsmine.mst.edu/the-mill/1>.